Exercise 5b Date:

Simulate the life cycle stages for UI design using the RAD model and develop a small interactive interface using OpenProj

AIM:

The aim is to recreate the lifecycle stages of UI design using the RAD model and design a small interactive interface with OpenProj

PROCEDURE:

Tool Link: https://sourceforge.net/projects/openproj/

DESCRIPTION

he Rapid Application Development (RAD) model was used to simulate the lifecycle stages of User Interface (UI) design using ProjectLibre.

In the **Requirements Planning phase**, key features such as user login and registration were identified. Use cases were defined to describe how users interact with the system, including logging in with valid credentials and creating a new account.

During the **User Design phase**, initial sketches of the login and registration screens were created. These were then converted into digital wireframes to represent the structure and layout of the interface.

In the **Rapid Prototyping phase**, interactive prototypes were developed based on the wireframes. These prototypes were tested, and feedback was collected to improve the design iteratively.

In the **User Acceptance and Testing phase**, the prototype was reviewed by users and stakeholders. Usability testing was conducted, and feedback was documented to ensure the interface met user requirements.

Finally, in the **Implementation phase**, the functional interface was developed based on the finalized design. Backend integration was considered for tasks such as user authentication.

All stages were planned and managed using ProjectLibre by creating tasks, assigning durations, and linking them with dependencies. The workflow was visualized using a Gantt

chart, clearly showing the sequence of activities.

Step 1: Requirements Planning

1. **Gather Requirements**
    - Identify key features and functionalities needed for your interface.
    - Example: A simple "Login" and "Register" interface with debug logs.
2. **Define Use Cases**

- Specify use cases for user login and registration.
- Example: User logs in with valid credentials, user registers with a new account.

Output in OpenProj:

- Create a new project.
- Add tasks: "Gather Requirements" and "Define Use Cases."
- Set durations and dependencies for each task.

Step 2: Wireframing

1. **Sketch Initial Designs**
    - Draw rough sketches of the "Login" and "Register" screens on paper.
2. **Create Digital Wireframes**

- Use a tool like Figma or Sketch to create digital wireframes.

Example Wireframes:

- **Login Screen:** Username field, Password field, Login button, Register link.
- **Register Screen:** Username field, Email field, Password field, Confirm Password field, Register button.

Output in OpenProj:

- Add tasks: "Sketch Initial Designs" and "Create Digital Wireframes."
- Allocate time and resources to complete these tasks.

Step 3: Rapid Prototyping

1. **Develop Prototypes**
    - Use a tool like Axure RP to convert wireframes into interactive prototypes.
2. **Test Prototypes**

- Share prototypes with stakeholders for feedback.
- Collect feedback and iterate on the design.

Output:

- Interactive prototypes for "Login" and "Register" screens.

Output in OpenProj:

- Add tasks: "Develop Prototypes" and "Test Prototypes."
- Set dependencies and milestones.

Step 4: User Acceptance / Testing

1. **Review Prototype**
    - Conduct user and stakeholder reviews.
2. **Conduct Usability Testing**

- Perform usability testing and document feedback.

Output:

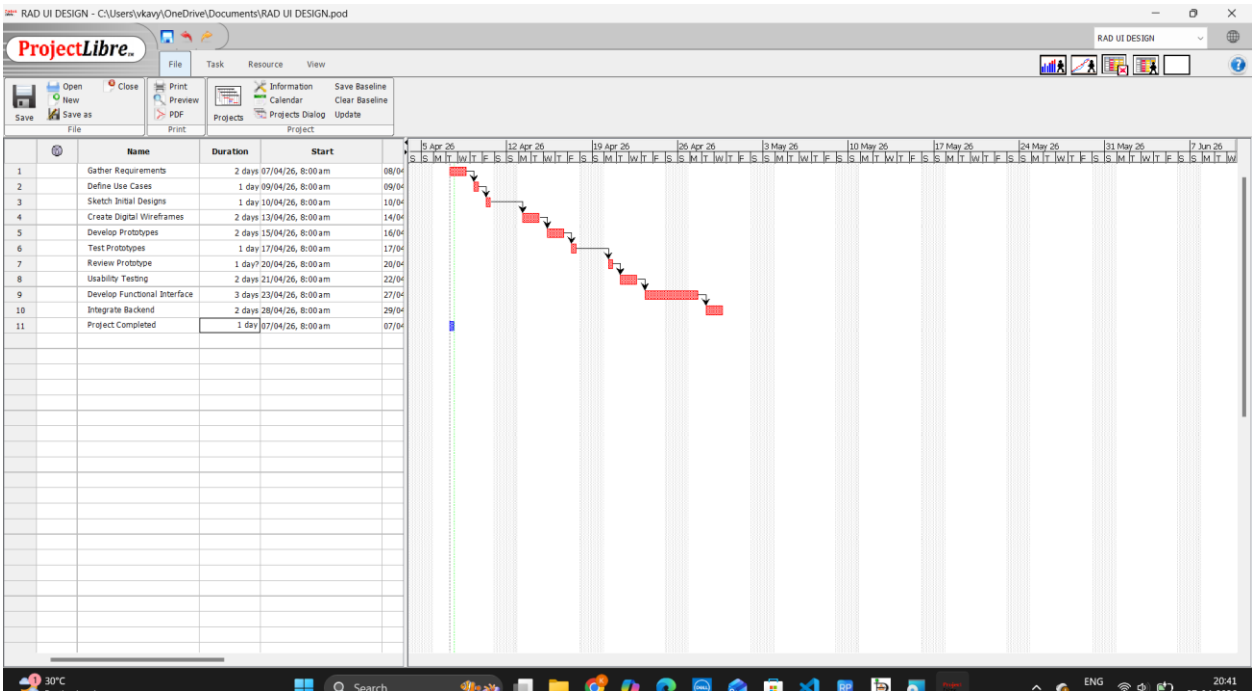
- Documented feedback and test results.

Output in OpenProj:

- Add tasks: "Review Prototype" and "Usability Testing."
- Track progress and resources.

Step 5: Implementation

1. **Develop Functional Interface**
    - Implement final designs and functionalities based on feedback.

2. **Integrate Backend (if required)**

- Connect the UI with backend services for tasks like user authentication.